2.72 3D Matrices: Eigenvalues and Eigenvectors

Worked Solutions

1 a)

$$\begin{pmatrix} 2 & 1 & 0 \\ 1 & 2 & 0 \\ 0 & 0 & 4 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} = \lambda \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 3 \\ 3 \\ 0 \end{pmatrix} = \lambda \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$$

$$\therefore \lambda = 3$$

1 b)

$$\begin{pmatrix} 1 & -2 & 1 \\ 1 & 1 & 1 \\ 1 & 2 & 2 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix} = \lambda \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 0 \\ 3 \\ 6 \end{pmatrix} = \lambda \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix}$$

$$\therefore \lambda = 3$$

1 c)

$$\begin{pmatrix} -7 & 1 & 0 \\ -16 & 0 & 2 \\ -7 & -1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix} = \lambda \begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} -4 \\ -12 \\ -8 \end{pmatrix} = \lambda \begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix}$$

$$\therefore \lambda = -4$$

2 a)

$$\begin{pmatrix} -11 & -6 & 10 \\ -4 & -9 & 8 \\ -10 & -12 & 14 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \lambda \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

$$\mathbf{A} - \lambda\mathbf{I} = \begin{pmatrix} -11 & -6 & 10 \\ -4 & -9 & 8 \\ -10 & -12 & 14 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} -11-\lambda & -6 & 10 \\ -4 & -9-\lambda & 8 \\ -10 & -12 & 14-\lambda \end{pmatrix}$$

$$\det(\mathbf{M}-\lambda\mathbf{I}) = \begin{vmatrix} -11-\lambda & -6 & 10 \\ -4 & -9-\lambda & 8 \\ -10 & -12 & 14-\lambda \end{vmatrix} = 0$$

$$\begin{vmatrix} -11-\lambda & -6 & -2\lambda-12 \\ -4 & -9-\lambda & 0 \\ -10 & -12 & -\lambda-6 \end{vmatrix} = 0$$

$$\begin{aligned} \det(\mathbf{M}-\lambda\mathbf{I}) &= (-2\lambda-12)(-10\lambda-42) + (-\lambda-6)(\lambda^2+20\lambda+75) \\ &= (\lambda+6)(20\lambda+84-\lambda^2-20\lambda-75) \\ &= (\lambda+6)(9-\lambda^2) \end{aligned}$$

$$\therefore (\lambda+6)(3-\lambda)(3+\lambda) = 0 \Rightarrow (\lambda-3)(\lambda+3)(\lambda+6) = 0 \quad \text{as req.}$$

2 b)

-6, -3 and 3

3 a)

$$\begin{pmatrix} 2 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & -3 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \lambda \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

$$\begin{aligned} \mathbf{A}-\lambda\mathbf{I} &= \begin{pmatrix} 2 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & -3 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} 2-\lambda & 1 & 1 \\ 0 & 1-\lambda & 1 \\ 0 & 0 & -3-\lambda \end{pmatrix} \end{aligned}$$

$$\det(\mathbf{A}-\lambda\mathbf{I}) = (2-\lambda)(1-\lambda)(-3-\lambda) - 0 + 0$$

So $(2-\lambda)(1-\lambda)(-3-\lambda) = 0$

$\therefore \lambda = 2,\ \lambda = 1$ or $\lambda = -3$

3 b)

$$\begin{pmatrix} 1 & -2 & 0 \\ 2 & 1 & 5 \\ 0 & 0 & 5 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \lambda \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

$$\begin{aligned} \mathbf{A} - \lambda\mathbf{I} &= \begin{pmatrix} 1 & -2 & 0 \\ 2 & 1 & 5 \\ 0 & 0 & 5 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} 1-\lambda & -2 & 0 \\ 2 & 1-\lambda & 5 \\ 0 & 0 & 5-\lambda \end{pmatrix} \end{aligned}$$

$$\det(\mathbf{A} - \lambda\mathbf{I}) = 0 - 0 + (5-\lambda)((1-\lambda)^2 + 4)$$

So $(5-\lambda)(\lambda^2 - 2\lambda + 5) = 0$

$\therefore \lambda = 5,\ \lambda = 1 \pm 2\mathrm{i}$

3 c)

$$\begin{pmatrix} 0 & -1 & 1 \\ -1 & -1 & 2 \\ -1 & 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \lambda \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

$$\begin{aligned} \mathbf{A} - \lambda\mathbf{I} &= \begin{pmatrix} 0 & -1 & 1 \\ -1 & -1 & 2 \\ -1 & 0 & 1 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} 0-\lambda & -1 & 1 \\ -1 & -1-\lambda & 2 \\ -1 & 0 & 1-\lambda \end{pmatrix} \end{aligned}$$

$$\begin{aligned} \det(\mathbf{A} - \lambda\mathbf{I}) &= -\lambda((-1-\lambda)(1-\lambda) - 0) + (-(1-\lambda) + 2) + (-1-\lambda) \\ &= -\lambda(\lambda^2 - 1) + \lambda + 1 - 1 - \lambda \end{aligned}$$

So $-\lambda(\lambda - 1)(\lambda + 1) = 0$

$\therefore \lambda = 0,\ \lambda = 1$ or $\lambda = -1$

3 d)

$$\begin{pmatrix} 1 & -1 & -3 \\ 0 & 4 & -8 \\ 0 & 4 & -4 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \lambda \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

$$\mathbf{A}-\lambda\mathbf{I}=\begin{pmatrix}1&-1&-3\\0&4&-8\\0&4&-4\end{pmatrix}-\lambda\begin{pmatrix}1&0&0\\0&1&0\\0&0&1\end{pmatrix}$$
$$=\begin{pmatrix}1-\lambda&-1&-3\\0&4-\lambda&-8\\0&4&-4-\lambda\end{pmatrix}$$

$$\det(\mathbf{A}-\lambda\mathbf{I})=(1-\lambda)\begin{vmatrix}4-\lambda&-8\\4&-4-\lambda\end{vmatrix}-0+0$$
$$=(1-\lambda)((4-\lambda)(-4-\lambda)+32)$$
$$=(1-\lambda)(\lambda^2-16+32)$$

So $(1-\lambda)(\lambda^2+16)=0$

$\therefore \lambda=1$ or $\lambda=\pm 4\mathrm{i}$

4)

$$\begin{pmatrix}p&-q&0\\q&p&0\\0&0&r\end{pmatrix}\begin{pmatrix}x\\y\\z\end{pmatrix}=\lambda\begin{pmatrix}x\\y\\z\end{pmatrix}$$

$$\mathbf{A}-\lambda\mathbf{I}=\begin{pmatrix}p&-q&0\\q&p&0\\0&0&r\end{pmatrix}-\lambda\begin{pmatrix}1&0&0\\0&1&0\\0&0&1\end{pmatrix}$$
$$=\begin{pmatrix}p-\lambda&-q&0\\q&p-\lambda&0\\0&0&r-\lambda\end{pmatrix}$$

$$\det(\mathbf{A}-\lambda\mathbf{I})=\begin{vmatrix}p-\lambda&-q&0\\q&p-\lambda&0\\0&0&r-\lambda\end{vmatrix}=0$$

$$\det(\mathbf{A}-\lambda\mathbf{I})=(r-\lambda)((p-\lambda)(p-\lambda)+q^2)$$
$$=(r-\lambda)(\lambda^2-2p\lambda+p^2+q^2)$$
$$\lambda^2-2p\lambda+p^2+q^2=0$$
$$\Rightarrow\lambda=\frac{2p\pm\sqrt{(-2p)^2-4(p^2+q^2)}}{2}$$
$$=\frac{2p\pm\sqrt{4p^2-4p^2-4q^2}}{2}$$
$$=\frac{2p\pm\sqrt{-4q^2}}{2}$$
$$=\frac{2p\pm 2\sqrt{-q^2}}{2}=p\pm q\mathrm{i}$$

$\therefore \lambda=r$ and $\lambda=p\pm q\mathrm{i}$

5 a)

$$\begin{pmatrix} 2 & 0 & -\frac{3}{2} \\ 0 & -5 & \frac{11}{2} \\ 0 & 0 & \frac{1}{2} \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \lambda \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

$$\begin{aligned} \mathbf{A} - \lambda\mathbf{I} &= \begin{pmatrix} 2 & 0 & -\frac{3}{2} \\ 0 & -5 & \frac{11}{2} \\ 0 & 0 & \frac{1}{2} \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} 2-\lambda & 0 & -\frac{3}{2} \\ 0 & -5-\lambda & \frac{11}{2} \\ 0 & 0 & \frac{1}{2}-\lambda \end{pmatrix} \end{aligned}$$

$$\det(\mathbf{A} - \lambda\mathbf{I}) = (2-\lambda)(-5-\lambda)\left(\frac{1}{2}-\lambda\right)$$

$\therefore \lambda = 2$, $\lambda = -5$ and $\lambda = \frac{1}{2}$

5 b)

$$\begin{aligned} \mathbf{M}^2 &= \begin{pmatrix} 2 & 0 & -\frac{3}{2} \\ 0 & -5 & \frac{11}{2} \\ 0 & 0 & \frac{1}{2} \end{pmatrix} \begin{pmatrix} 2 & 0 & -\frac{3}{2} \\ 0 & -5 & \frac{11}{2} \\ 0 & 0 & \frac{1}{2} \end{pmatrix} \\ &= \begin{pmatrix} 4 & 0 & -\frac{15}{4} \\ 0 & 25 & -\frac{99}{4} \\ 0 & 0 & \frac{1}{4} \end{pmatrix} \end{aligned}$$

$$\begin{pmatrix} 4 & 0 & -\frac{15}{4} \\ 0 & 25 & -\frac{99}{4} \\ 0 & 0 & \frac{1}{4} \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \lambda \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

$$\begin{aligned} \mathbf{A} - \lambda\mathbf{I} &= \begin{pmatrix} 4 & 0 & -\frac{15}{4} \\ 0 & 25 & -\frac{99}{4} \\ 0 & 0 & \frac{1}{4} \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} 4-\lambda & 0 & -\frac{15}{4} \\ 0 & 25-\lambda & -\frac{99}{4} \\ 0 & 0 & \frac{1}{4}-\lambda \end{pmatrix} \end{aligned}$$

$$\det(\mathbf{A} - \lambda\mathbf{I}) = (4-\lambda)(25-\lambda)\left(\frac{1}{4}-\lambda\right)$$

$\therefore \lambda = 4$, $\lambda = 25$ and $\lambda = \frac{1}{4}$

5 c)

The eigenvalues of $\mathbf{A}$ are $\lambda = 2$, $\lambda = -5$ and $\lambda = \frac{1}{2}$

$\Rightarrow \lambda^2 = 4$, $\lambda^2 = 25$ and $\lambda^2 = \frac{1}{4}$ which are the eigenvalues of $\mathbf{A}^2$

5 d)

If $\mathbf{Mv} = \lambda\mathbf{v}$, then

$$\mathbf{M}^2\mathbf{v} = \mathbf{M}(\mathbf{Mv}) = \mathbf{M}(\lambda\mathbf{v}) = \lambda\mathbf{Mv} = \lambda(\lambda\mathbf{v}) = \lambda^2\mathbf{v} \quad \text{as req.}$$

6 a)

$$\begin{pmatrix} 3 & -3 & 1 \\ 0 & -5 & 6 \\ 0 & -3 & 4 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \lambda \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

$$\begin{aligned} \mathbf{A} - \lambda\mathbf{I} &= \begin{pmatrix} 3 & -3 & 1 \\ 0 & -5 & 6 \\ 0 & -3 & 4 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} 3-\lambda & -3 & 1 \\ 0 & -5-\lambda & 6 \\ 0 & -3 & 4-\lambda \end{pmatrix} \end{aligned}$$

$$\det(\mathbf{A} - \lambda\mathbf{I}) = \begin{vmatrix} 3-\lambda & -3 & 1 \\ 0 & -5-\lambda & 6 \\ 0 & -3 & 4-\lambda \end{vmatrix} = 0$$

$$\begin{aligned} &= (3-\lambda)((-5-\lambda)(4-\lambda)+18) \\ &= (3-\lambda)(\lambda^2+\lambda-2) \\ &= (3-\lambda)(\lambda+2)(\lambda-1) \end{aligned}$$

$\therefore \lambda = -2$, $\lambda = 1$ and $\lambda = 3$

6 b)

$$\begin{aligned} \mathbf{M}^3 &= \begin{pmatrix} 3 & -3 & 1 \\ 0 & -5 & 6 \\ 0 & -3 & 4 \end{pmatrix} \begin{pmatrix} 3 & -3 & 1 \\ 0 & -5 & 6 \\ 0 & -3 & 4 \end{pmatrix} \begin{pmatrix} 3 & -3 & 1 \\ 0 & -5 & 6 \\ 0 & -3 & 4 \end{pmatrix} \\ &= \begin{pmatrix} 3 & -3 & 1 \\ 0 & -5 & 6 \\ 0 & -3 & 4 \end{pmatrix} \begin{pmatrix} 9 & 3 & -11 \\ 0 & 7 & -6 \\ 0 & 3 & -2 \end{pmatrix} \\ &= \begin{pmatrix} 27 & -9 & -17 \\ 0 & -17 & 18 \\ 0 & -9 & 10 \end{pmatrix} \end{aligned}$$

$$\begin{pmatrix} 27 & -9 & -17 \\ 0 & -17 & 18 \\ 0 & -9 & 10 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \lambda \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

$$\begin{aligned} \mathbf{A} - \lambda\mathbf{I} &= \begin{pmatrix} 27 & -9 & -17 \\ 0 & -17 & 18 \\ 0 & -9 & 10 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} 27-\lambda & -9 & -17 \\ 0 & -17-\lambda & 18 \\ 0 & -9 & 10-\lambda \end{pmatrix} \end{aligned}$$

$$\det(\mathbf{A} - \lambda\mathbf{I}) = \begin{vmatrix} 27-\lambda & -9 & -17 \\ 0 & -17-\lambda & 18 \\ 0 & -9 & 10-\lambda \end{vmatrix} = 0$$

$$\begin{aligned} &= (27-\lambda)((-17-\lambda)(10-\lambda)+162) \\ &= (27-\lambda)(\lambda^2+7\lambda-8) \\ &= (27-\lambda)(\lambda+8)(\lambda-1) \end{aligned}$$

$\therefore \lambda = -8$, $\lambda = 1$ and $\lambda = 27$

6 c)

The eigenvalues of $\mathbf{A}$ are $\lambda = -2$, $\lambda = 1$ and $\lambda = 3$

$\Rightarrow \lambda^3 = -8$, $\lambda^3 = 1$ and $\lambda^3 = 27$ which are the eigenvalues of $\mathbf{A}^3$

6 d)

If $\mathbf{Mv} = \lambda\mathbf{v}$, then

$$\begin{aligned} \mathbf{M}^2\mathbf{v} &= \mathbf{M}(\mathbf{Mv}) = \mathbf{M}(\lambda\mathbf{v}) = \lambda\mathbf{Mv} = \lambda(\lambda\mathbf{v}) = \lambda^2\mathbf{v} \\ \Rightarrow \mathbf{M}^3\mathbf{v} &= \mathbf{M}(\mathbf{M}^2\mathbf{v}) = \mathbf{M}(\lambda^2\mathbf{v}) = \lambda^2(\mathbf{Mv}) = \lambda^2(\lambda\mathbf{v}) = \lambda^3\mathbf{v} \quad \text{as req.} \end{aligned}$$

7 a)

$$\begin{pmatrix} -5 & 2 & 2 \\ -18 & 7 & 6 \\ -6 & 2 & 3 \end{pmatrix} \begin{pmatrix} -5 & 2 & 2 \\ -18 & 7 & 6 \\ -6 & 2 & 3 \end{pmatrix} + 3 \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} -23 & 8 & 8 \\ -72 & 25 & 24 \\ -24 & 8 & 9 \end{pmatrix} + 3 \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} -20 & 8 & 8 \\ -72 & 28 & 24 \\ -24 & 8 & 12 \end{pmatrix}$$

$$k \begin{pmatrix} -20 & 8 & 8 \\ -72 & 28 & 24 \\ -24 & 8 & 12 \end{pmatrix} = 4 \begin{pmatrix} -5 & 2 & 2 \\ -18 & 7 & 6 \\ -6 & 2 & 3 \end{pmatrix}$$

$$\therefore k = 4$$

7 b)

$$\begin{aligned}
\mathbf{M}^2 + 3\mathbf{I} &= 4\mathbf{M} \\
\mathbf{M}^2 - 4\mathbf{M} + 3\mathbf{I} &= \mathbf{0} \\
\mathbf{M} - 4\mathbf{I} + 3\mathbf{M}^{-1} &= \mathbf{0} \\
3\mathbf{M}^{-1} &= 4\mathbf{I} - \mathbf{M} \\
\mathbf{M}^{-1} &= \frac{4}{3}\mathbf{I} - \frac{1}{3}\mathbf{M} \\
&= -\frac{1}{3}\mathbf{M} + \frac{4}{3}\mathbf{I} \quad \text{as req.}
\end{aligned}$$

7 c)

$$\begin{pmatrix} -5 & 2 & 2 \\ -18 & 7 & 6 \\ -6 & 2 & 3 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \lambda \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

$$\begin{aligned}
\mathbf{M} - \lambda\mathbf{I} &= \begin{pmatrix} -5 & 2 & 2 \\ -18 & 7 & 6 \\ -6 & 2 & 3 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \\
&= \begin{pmatrix} -5-\lambda & 2 & 2 \\ -18 & 7-\lambda & 6 \\ -6 & 2 & 3-\lambda \end{pmatrix}
\end{aligned}$$

$$\det(\mathbf{M} - \lambda\mathbf{I}) = \begin{vmatrix} -5-\lambda & 2 & 2 \\ -18 & 7-\lambda & 6 \\ -6 & 2 & 3-\lambda \end{vmatrix} = 0$$

$$\begin{vmatrix} -5-\lambda & 0 & 2 \\ -18 & 1-\lambda & 6 \\ -6 & -1+\lambda & 3-\lambda \end{vmatrix} = 0$$

$$\begin{vmatrix} -5-\lambda & 0 & 2 \\ -24 & 0 & 9-\lambda \\ -6 & -1+\lambda & 3-\lambda \end{vmatrix} = 0$$

$$\begin{aligned}
&= -(-1+\lambda)((-5-\lambda)(9-\lambda) - (2)(-24)) \\
&= -(-1+\lambda)(\lambda^2 - 4\lambda - 45 + 48) \\
&= -(-1+\lambda)(\lambda-1)(\lambda-3)
\end{aligned}$$

$\therefore \lambda = 1$, $\lambda = 1$ and $\lambda = 3$

7 d)

$1^2 + 3 = 4 = 4 \times 1$ LHS=RHS $\therefore$ true

$3^2 + 3 = 12 = 4 \times 3$ LHS=RHS $\therefore$ true

8 a)

$$\begin{pmatrix} 1 & 3 & -1 \\ -2 & 3 & 2 \\ 1 & 0 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \lambda \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

$$\begin{aligned} \mathbf{M} - \lambda\mathbf{I} &= \begin{pmatrix} 1 & 3 & -1 \\ -2 & 3 & 2 \\ 1 & 0 & 2 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} 1-\lambda & 3 & -1 \\ -2 & 3-\lambda & 2 \\ 1 & 0 & 2-\lambda \end{pmatrix} \end{aligned}$$

$$\det(\mathbf{M} - \lambda\mathbf{I}) = \begin{vmatrix} 1-\lambda & 3 & -1 \\ -2 & 3-\lambda & 2 \\ 1 & 0 & 2-\lambda \end{vmatrix} = 0$$

$$\begin{vmatrix} 1-\lambda & 3 & -\lambda \\ -2 & 3-\lambda & 0 \\ 1 & 0 & 3-\lambda \end{vmatrix} = 0$$

$$\begin{vmatrix} -\lambda & 3 & -3 \\ -2 & 3-\lambda & 0 \\ 1 & 0 & 3-\lambda \end{vmatrix} = 0$$

$$\begin{vmatrix} -\lambda & 3 & -3 \\ -2 & 3-\lambda & 0 \\ 1 & 0 & 3-\lambda \end{vmatrix} = 0$$

$$\begin{aligned} \det(\mathbf{M} - \lambda\mathbf{I}) &= -3(\lambda - 3) - 0 + (3-\lambda)(\lambda^2 - 3\lambda + 6) \\ &= (3-\lambda)(3 + \lambda^2 - 3\lambda + 6) \end{aligned}$$

$\therefore (\lambda - 3)(\lambda^2 - 3\lambda + 9) = 0$ as req.

8 b)

$$\lambda = 3 \text{ and } \lambda = \frac{3}{2} \pm \frac{3\sqrt{3}}{2}\mathrm{i}$$

When $\lambda = 3$ then $\mathbf{M} - 3\mathbf{I} = \begin{pmatrix} -2 & 3 & -1 \\ -2 & 0 & 2 \\ 1 & 0 & -1 \end{pmatrix}$

$$\begin{pmatrix} -2 & 3 & -1 \\ -2 & 0 & 2 \\ 1 & 0 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0$$

$\Rightarrow -2x + 3y - z = 0$ and $-2x + 2z = 0$ and $-x - z = 0$

So $x = z$ and so $y = x = z$

$\therefore$ when $\lambda = 3$, $\mathbf{v} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$

9 a)

When $\lambda = 2$ then $\mathbf{A} - 2\mathbf{I} = \begin{pmatrix} 4 & 3 & 6 \\ -2 & -5 & -10 \\ -8 & -3 & -6 \end{pmatrix}$

$$\begin{pmatrix} 4 & 3 & 6 \\ -2 & -5 & -10 \\ -8 & -3 & -6 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0$$

$$\begin{aligned} \Rightarrow 4x + 3y + 6z &= 0 \quad (1) \\ -2x - 5y - 10z &= 0 \quad (2) \\ -8x - 3y - 6z &= 0 \quad (3) \end{aligned}$$

(1) $+2imes(2)$:

$$-7y - 14z = 0 \quad \therefore y = -2z$$

Sub into (1):

$$4x = 0 \quad \therefore x = 0$$

$\therefore$ when $\lambda = 2$, $\mathbf{v} = \begin{pmatrix} 0 \\ -2 \\ 1 \end{pmatrix}$

9 b)

When $\lambda = 0$ then $\mathbf{A} - 0\mathbf{I} = \begin{pmatrix} -2 & 5 & 7 \\ 1 & 0 & -1 \\ -1 & 1 & 2 \end{pmatrix}$

$$\begin{pmatrix} -2 & 5 & 7 \\ 1 & 0 & -1 \\ -1 & 1 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0$$

$$\begin{aligned} \Rightarrow -2x + 5y + 7z &= 0 \quad (1) \\ x - z &= 0 \quad (2) \\ -x + y + 2z &= 0 \quad (3) \end{aligned}$$

From (2):

$$x = z$$

Sub into (3):

$$y + x = 0 \quad \therefore -y = x$$

$\therefore$ when $\lambda = 0$, $\mathbf{v} = \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}$

9 c)

When $\lambda = 1$ then $\mathbf{A} + 1\mathbf{I} = \begin{pmatrix} 5 & -8 & 3 \\ 3 & -6 & 3 \\ 4 & -10 & 6 \end{pmatrix}$

$$\begin{pmatrix} 5 & -8 & 3 \\ 3 & -6 & 3 \\ 4 & -10 & 6 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0$$

$$\begin{aligned} \Rightarrow 5x - 8y + 3z = 0 \quad (1) \\ 3x - 6y + 3z = 0 \quad (2) \\ 4x - 10y + 6z = 0 \quad (3) \end{aligned}$$

(2) $-$(1):

$$2y - 2x = 0 \quad \therefore y = x$$

Sub into (1):

$$-3x + 3z = 0 \quad \therefore z = x$$

$\therefore$ when $\lambda = 1$, $\mathbf{v} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$

9 d)

When $\lambda = 3$ then $\mathbf{A} - 3\mathbf{I} = \begin{pmatrix} -1 & 1 & 2 \\ 4 & 2 & -1 \\ -4 & 1 & 3 \end{pmatrix}$

$$\begin{pmatrix} -4 & 1 & 2 \\ 4 & -1 & -1 \\ -4 & 1 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0$$

$$\Rightarrow -4x + y + 2z = 0 \quad (1)$$
$$4x - y - z = 0 \quad (2)$$
$$-4x + y = 0 \quad (3)$$

From (3):

$$y = 4x$$

Sub into (1):

$$2z = 0 \quad \therefore z = 0$$

$\therefore$ when $\lambda = 3$, $\mathbf{v} = \begin{pmatrix} 1 \\ 4 \\ 0 \end{pmatrix}$

9 e)

When $\lambda = 3$ then $\mathbf{A} + 5\mathbf{I} = \begin{pmatrix} 0 & 1 & 1 \\ \frac{3}{2} & 5 & 2 \\ -1 & 3 & 5 \end{pmatrix}$

$$\begin{pmatrix} 0 & 1 & 1 \\ \frac{3}{2} & 5 & 2 \\ -1 & 3 & 5 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0$$

$$\Rightarrow y + z = 0 \quad (1)$$
$$\frac{3}{2}x + 5y + 2z = 0 \quad (2)$$
$$-x + 3y + 5z = 0 \quad (3)$$

From (1):

$$y = -z$$

Sub into (3):

$$-x + 2z = 0 \quad \therefore 2z = x$$

$\therefore$ when $\lambda = 3$, $\mathbf{v} = \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix}$

10 a)

$$\begin{pmatrix} 4 & 1 & 0 \\ 1 & 4 & 0 \\ 1 & 1 & 5 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \lambda \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

$$\begin{aligned} \mathbf{M} - \lambda\mathbf{I} &= \begin{pmatrix} 4 & 1 & 0 \\ 1 & 4 & 0 \\ 1 & 1 & 5 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} 4-\lambda & 1 & 0 \\ 1 & 4-\lambda & 0 \\ 1 & 1 & 5-\lambda \end{pmatrix} \end{aligned}$$

$$\begin{aligned} \det(\mathbf{M} - \lambda\mathbf{I}) &= 0 - 0 - (5-\lambda)\begin{vmatrix} 4-\lambda & 1 \\ 1 & 4-\lambda \end{vmatrix} \\ &= (\lambda - 5)(\lambda^2 - 8\lambda + 15) \end{aligned}$$

So $(\lambda - 5)(\lambda - 5)(\lambda - 3) = 0$

So $\lambda = 5$ is a repeated eigenvalue of $\mathbf{M}$

10 b)

$\lambda_1 = 5$, $\lambda_2 = 5$ or $\lambda_3 = 3$

10 c)

When $\lambda_1 = 5$ then $\mathbf{A} - 5\mathbf{I} = \begin{pmatrix} -1 & 1 & 0 \\ 1 & -1 & 0 \\ 1 & 1 & 0 \end{pmatrix}$

$$\begin{pmatrix} -1 & 1 & 0 \\ 1 & -1 & 0 \\ 1 & 1 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0$$

$\Rightarrow -x + y = 0$ and $x + y = 0$

So $x = 0$ and $y = 0$ and z may take any value

$\therefore$ when $\lambda_1 = 5$, $\mathbf{v}_1 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$

When $\lambda_3 = 3$ then $\mathbf{M} - 3\mathbf{I} = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 2 \end{pmatrix}$

$$\begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0$$

$\Rightarrow x + y = 0$ and $x + y + 2z = 0$

So $y = -x$ and $z = 0$ and z may take any value

$\therefore$ when $\lambda_3 = 3$, $\mathbf{v}_3 = \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}$

$\therefore \lambda_1 = 5$, $\mathbf{v}_1 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$ and $\lambda_3 = 3$, $\mathbf{v}_3 = \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}$

11 a)

$$\begin{pmatrix} 6 & 1 & -4 \\ 3 & 4 & -4 \\ 7 & 1 & -5 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \lambda \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

$$\begin{aligned} \mathbf{M} - \lambda\mathbf{I} &= \begin{pmatrix} 6 & 1 & -4 \\ 3 & 4 & -4 \\ 7 & 1 & -5 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} 6-\lambda & 1 & -4 \\ 3 & 4-\lambda & -4 \\ 7 & 1 & -5-\lambda \end{pmatrix} \end{aligned}$$

$$\det(\mathbf{M} - \lambda\mathbf{I}) = \begin{vmatrix} 6-\lambda & 1 & -4 \\ 3 & 4-\lambda & -4 \\ 7 & 1 & -5-\lambda \end{vmatrix} = 0$$

$$\begin{vmatrix} 3-\lambda & -3+\lambda & 0 \\ 3 & 4-\lambda & -4 \\ 7 & 1 & -5-\lambda \end{vmatrix} = 0$$

$$\begin{vmatrix} 3-\lambda & 0 & 0 \\ 3 & 7-\lambda & -4 \\ 7 & 8 & -5-\lambda \end{vmatrix} = 0$$

$$\begin{aligned} &= (3-\lambda) \begin{vmatrix} 7-\lambda & -4 \\ 8 & -5-\lambda \end{vmatrix} \\ &= (3-\lambda)(\lambda^2 - 2\lambda - 3) \end{aligned}$$

So $\lambda = 3$ is an eigenvalue of $\mathbf{M}$

11 b)

$$(\lambda - 3)(\lambda^2 - 2\lambda - 3) = (\lambda - 3)(\lambda - 3)(\lambda + 1)$$

$\lambda_1 = 3$, $\lambda_2 = 3$ or $\lambda_3 = -1$

11 c)

When $\lambda_1 = 3$ then $\mathbf{M} - 3\mathbf{I} = \begin{pmatrix} 3 & 1 & -4 \\ 3 & 1 & -4 \\ 7 & 1 & -8 \end{pmatrix}$

$$\begin{pmatrix} 3 & 1 & -4 \\ 3 & 1 & -4 \\ 7 & 1 & -8 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0$$

$$\begin{aligned} \Rightarrow 3x + y - 4z &= 0 \quad (1) \\ 7x + y - 8z &= 0 \quad (2) \end{aligned}$$

(2) $-$(1): $4x - 4z = 0 \ \therefore x = z$

Sub into (1): $-x + y = 0 \ \therefore x = y$

$\therefore$ when $\lambda_1 = 3$, $\mathbf{v}_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$

When $\lambda_3 = -1$ then $\mathbf{A} + 1\mathbf{I} = \begin{pmatrix} 7 & 1 & -4 \\ 3 & 5 & -4 \\ 7 & 1 & -4 \end{pmatrix}$

$$\begin{pmatrix} 7 & 1 & -4 \\ 3 & 5 & -4 \\ 7 & 1 & -4 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0$$

$$\begin{aligned} \Rightarrow 7x + y - 4z &= 0 \quad (1) \\ 3x + 5y - 4z &= 0 \quad (2) \end{aligned}$$

(2) $-$(1): $-4x + 4y = 0 \ \therefore y = x$

Sub into (1): $8x - 4z = 0 \ \therefore z = 2x$

$\therefore$ when $\lambda_3 = -1$, $\mathbf{v}_3 = \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}$

$\therefore$ when $\lambda_1 = 3$, $\mathbf{v}_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$, $\lambda_3 = -1$, $\mathbf{v}_3 = \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}$

12 a)

$$\begin{pmatrix} -4 & 16 & -8 \\ -5 & 13 & -6 \\ -10 & 22 & -10 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \lambda \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

$$\mathbf{M} - \lambda\mathbf{I} = \begin{pmatrix} -4 & 16 & -8 \\ -5 & 13 & -6 \\ -10 & 22 & -10 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} -4-\lambda & 16 & -8 \\ -5 & 13-\lambda & -6 \\ -10 & 22 & -10-\lambda \end{pmatrix}$$

$$\det(\mathbf{M}-\lambda\mathbf{I}) = \begin{vmatrix} -4-\lambda & 16 & -8 \\ -5 & 13-\lambda & -6 \\ -10 & 22 & -10-\lambda \end{vmatrix} = 0$$

$$\begin{vmatrix} -4-\lambda & 0 & -8 \\ -5 & 1-\lambda & -6 \\ -10 & 2-2\lambda & -10-\lambda \end{vmatrix} = 0$$

$$\begin{vmatrix} -4-\lambda & 0 & -8 \\ -5 & 1-\lambda & -6 \\ 0 & 0 & 2-\lambda \end{vmatrix} = 0$$

$$= (2-\lambda)(-4-\lambda)(1-\lambda)$$

So $\lambda = 2$ is an eigenvalue of $\mathbf{M}$

12 b)

$$(2-\lambda)(-4-\lambda)(1-\lambda) = 0$$

$\lambda_1 = 2$, $\lambda_2 = -4$ or $\lambda_3 = 1$

12 c)

When $\lambda_1 = 2$ then $\mathbf{A} - 2\mathbf{I} = \begin{pmatrix} -6 & 16 & -8 \\ -5 & 11 & -6 \\ -10 & 22 & -12 \end{pmatrix}$

$$\begin{pmatrix} -6 & 16 & -8 \\ -5 & 11 & -6 \\ -10 & 22 & -12 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0$$

$$\Rightarrow -6x + 16y - 8z = 0 \quad (1)$$
$$-5x + 11y - 6z = 0 \quad (2)$$
$$5 \times (1) \Rightarrow -30x + 80y - 40z = 0 \quad (3)$$
$$6 \times (2) \Rightarrow -30x + 66y - 36z = 0 \quad (4)$$

(3) $-$(4): $14y - 4z = 0 \ \therefore 7y = 2z$

Sub into (1): $-6x + 16y - 28y = 0 \Rightarrow -6x - 12y = 0 \ \therefore -2y = x$

$\therefore$ when $\lambda_1 = 2$, $\mathbf{v}_1 = \begin{pmatrix} 4 \\ -2 \\ -7 \end{pmatrix}$

When $\lambda_2 = -4$ then $\mathbf{A} + 4\mathbf{I} = \begin{pmatrix} 0 & 16 & -8 \\ -5 & 17 & -6 \\ -10 & 22 & -6 \end{pmatrix}$

$$\begin{pmatrix} 0 & 16 & -8 \\ -5 & 17 & -6 \\ -10 & 22 & -6 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0$$

$$\begin{aligned} \Rightarrow 16y - 8z = 0 \quad (1) \\ -5x + 17y - 6z = 0 \quad (2) \\ -10x + 22y - 6z = 0 \quad (3) \end{aligned}$$

From (1): $2y = z$

Sub into (2): $-5x + 17y - 12y = 0 \Rightarrow -5x + 5y = 0 \ \therefore y = x$

$\therefore$ when $\lambda_2 = -4$, $\mathbf{v}_2 = \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}$

When $\lambda_3 = 1$ then $\mathbf{A} - 1\mathbf{I} = \begin{pmatrix} -5 & 16 & -8 \\ -5 & 12 & -6 \\ -10 & 22 & -11 \end{pmatrix}$

$$\begin{pmatrix} -5 & 16 & -8 \\ -5 & 12 & -6 \\ -10 & 22 & -11 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0$$

$$\begin{aligned} \Rightarrow -5x + 16y - 8z = 0 \quad (1) \\ -5x + 12y - 6z = 0 \quad (2) \end{aligned}$$

(1) $-$(2): $4y - 2z = 0 \ \therefore z = 2y$

Sub into (1): $-5x = 0 \ \therefore x = 0$

$\therefore$ when $\lambda_3 = 1$, $\mathbf{v}_2 = \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix}$

$\therefore$ when $\lambda_1 = 2$, $\mathbf{v}_1 = \begin{pmatrix} 4 \\ -2 \\ -7 \end{pmatrix}$, $\lambda_2 = -4$, $\mathbf{v}_2 = \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}$, $\lambda_3 = 1$, $\mathbf{v}_2 = \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix}$

13)

Since $\mathbf{Mv} = \lambda\mathbf{v}$

$$\mathbf{Mv} = \begin{pmatrix} -1 & p & 0 \\ p & 0 & 2 \\ 0 & 2 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \\ q \end{pmatrix} = \begin{pmatrix} -1 + 2p \\ p + 4 \\ 4 + q \end{pmatrix}$$

$$\lambda\mathbf{v} = 3 \begin{pmatrix} 1 \\ 2 \\ q \end{pmatrix} = \begin{pmatrix} 3 \\ 6 \\ 3q \end{pmatrix}$$

$-1 + 2p = 3 \ \therefore p = 2$

$4 + q = 3q \ \therefore q = 2$

14 a)

$$\frac{1}{14}\begin{pmatrix}10 & -44 & -2k\\ 0 & 42 & 0\\ 6 & -53 & -17\end{pmatrix}\begin{pmatrix}1\\0\\2\end{pmatrix}=\frac{1}{7}\begin{pmatrix}5-2k\\0\\-14\end{pmatrix}$$
$$\Rightarrow \lambda_3\begin{pmatrix}1\\0\\2\end{pmatrix}=\frac{1}{7}\begin{pmatrix}5-2k\\0\\-14\end{pmatrix}$$

$2\lambda_3 = -2 \ \therefore \lambda_3 = -1$
$-1 = \frac{5-2k}{7} \Rightarrow -12 = -2k \Rightarrow \therefore k = 6$

14 b)

$$\frac{1}{14}\begin{pmatrix}10 & -44 & -12\\ 0 & 42 & 0\\ 6 & -53 & -17\end{pmatrix}\begin{pmatrix}a\\0\\1\end{pmatrix}=\frac{1}{14}\begin{pmatrix}10a-12\\0\\6a-17\end{pmatrix}$$
$$\Rightarrow \lambda_2\begin{pmatrix}a\\0\\1\end{pmatrix}=\frac{1}{14}\begin{pmatrix}10a-12\\0\\6a-17\end{pmatrix}$$

$14a\lambda_2 = 10a - 12$ and $\lambda_2 = \frac{6a-17}{14}$
So that $14a\left(\frac{6a-17}{14}\right) = 10a - 12 \Rightarrow 6a^2 - 17a = 10a - 12 \Rightarrow 6a^2 - 27a + 12 = 0$
$a = \frac{1}{2}$ or $a = 4$ but a is an integer.
$\therefore a = 4$

14 c)

$$\lambda_2 = \frac{6a-17}{14} = \frac{24-17}{14} = \frac{1}{2}$$

$\therefore \lambda_1 = 3$, $\lambda_2 = \frac{1}{2}$, $\lambda_3 = -1$

15 a)

$$\begin{pmatrix}6 & -2 & 2\\ 2-k & 2+k & k-2\\ 4-k & k-4 & k+4\end{pmatrix}\begin{pmatrix}x\\y\\z\end{pmatrix}=\lambda\begin{pmatrix}x\\y\\z\end{pmatrix}$$

$$\mathbf{M}-\lambda\mathbf{I}=\begin{pmatrix}6 & -2 & 2\\ 2-k & 2+k & k-2\\ 4-k & k-4 & k+4\end{pmatrix}-\lambda\begin{pmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{pmatrix}$$
$$=\begin{pmatrix}6-\lambda & -2 & 2\\ 2-k & 2+k-\lambda & k-2\\ 4-k & k-4 & k+4-\lambda\end{pmatrix}$$

$$\det(\mathbf{M}-\lambda\mathbf{I}) = \begin{vmatrix} 6-\lambda & -2 & 2 \\ 2-k & 2+k-\lambda & k-2 \\ 4-k & k-4 & k+4-\lambda \end{vmatrix} = 0$$

$$\begin{vmatrix} 6-\lambda & -2 & 0 \\ 2-k & 2+k-\lambda & 2k-\lambda \\ 4-k & k-4 & 2k-\lambda \end{vmatrix} = 0$$

$$\begin{vmatrix} 6-\lambda & -2 & 0 \\ -2 & 6-\lambda & 0 \\ 4-k & k-4 & 2k-\lambda \end{vmatrix} = 0$$

$$= (2k-\lambda)\begin{vmatrix} 6-\lambda & -2 \\ -2 & 6-\lambda \end{vmatrix}$$

$$= (2k-\lambda)(\lambda^2-12\lambda+32) = (2k-\lambda)(\lambda-4)(\lambda-8)$$

So $\lambda = 2k$ is an eigenvalue of $\mathbf{M}$

15 b)

When $\lambda_3 = 2k$ then $\mathbf{M}-3\mathbf{I} = \begin{pmatrix} 6-2k & -2 & 2 \\ 2-k & 2-k & k-2 \\ 4-k & k-4 & 4-k \end{pmatrix}$

$$\begin{pmatrix} 6-2k & -2 & 2 \\ 2-k & 2-k & k-2 \\ 4-k & k-4 & 4-k \end{pmatrix}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0$$

$$\Rightarrow (6-2k)x - 2y + 2z = 0 \quad (1)$$
$$(2-k)x + (2-k)y + (k-2)z = 0 \quad (2)$$
$$(4-k)x + (k-4)y + (4-k)z = 0 \quad (3)$$

From (2): $(2-k)(x+y-z) = 0$

$$\Rightarrow x + y - z = 0$$

From (3): $(4-k)(x-y+z) = 0$

$$\Rightarrow x - y + z = 0$$

$\therefore x = 0$ and $y = z$

$\therefore$ when $\lambda_3 = 2k$, $\mathbf{v}_1 = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$

16 a)

$$\begin{pmatrix} k-2 & -3 & 3 \\ 6 & -5 & 3 \\ 0 & 0 & -2 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \lambda \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

$$\begin{aligned} \mathbf{M} - \lambda\mathbf{I} &= \begin{pmatrix} k-2 & -3 & 3 \\ 6 & -5 & 3 \\ 0 & 0 & -2 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} k-2-\lambda & -3 & 3 \\ 6 & -5-\lambda & 3 \\ 0 & 0 & -2-\lambda \end{pmatrix} \end{aligned}$$

$$\det(\mathbf{M} - \lambda\mathbf{I}) = \begin{vmatrix} k-2-\lambda & -3 & 3 \\ 6 & -5-\lambda & 3 \\ 0 & 0 & -2-\lambda \end{vmatrix} = 0$$

$$\begin{aligned} &= (-2-\lambda) \begin{vmatrix} k-2-\lambda & -3 \\ 6 & -5-\lambda \end{vmatrix} \\ &= -(\lambda+2)((k-2-\lambda)(-5-\lambda) - (-18)) \\ &= -(\lambda+2)(\lambda^2 + (7-k)\lambda + 28 - 5k) = 0 \end{aligned}$$

So consider the quadratic: $b^2 - 4ac = 0$ creates a repeated root

$$\begin{aligned} (7-k)^2 - 4(28-5k) = 0 &\Rightarrow k^2 - 14k + 49 + 20k - 112 = 0 \\ &\Rightarrow k^2 + 6k - 63 = 0 \end{aligned}$$

so $k = -3 \pm 6\sqrt{2}$ which is not an integer.

So $\lambda = -2$ must be the repeated root.

When $\lambda = -2$ then

$$\begin{aligned} 4 - 2(7-k) + 28 - 5k = 0 &\Rightarrow 4 - 14 + 2k + 28 - 5k = 0 \\ &\therefore k = 6 \end{aligned}$$

16 b)

When $k = 6$

$$-(\lambda+2)(\lambda^2 + \lambda - 2) = -(\lambda+2)(\lambda+2)(\lambda-1)$$

So eigenvalues are $\lambda_1 = -2$, $\lambda_2 = -2$ and $\lambda_3 = 1$

When $\lambda_1 = -2$ then $\mathbf{A} + 2\mathbf{I} = \begin{pmatrix} 6 & -3 & 3 \\ 6 & -3 & 3 \\ 0 & 0 & 0 \end{pmatrix}$

$$\begin{pmatrix} 6 & -3 & 3 \\ 6 & -3 & 3 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0$$

$$\Rightarrow 6x - 3y + 3z = 0 \quad (1)$$

So $x = 0$ gives $\begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$ and $y = 0$ gives $\begin{pmatrix} 1 \\ 0 \\ -2 \end{pmatrix}$ and $z = 0$ gives $\begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}$

Any two linearly independent ones span the eigenspace for $\lambda = -2$

Since $\begin{pmatrix} 1 \\ 0 \\ -2 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} - 2\begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$ then $\begin{pmatrix} 1 \\ 0 \\ -2 \end{pmatrix}$ is dependent. So choose $\begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$ and $\begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}$

$\therefore$ when $\lambda_1 = -2$, $\mathbf{v}_1 = \begin{pmatrix} 1 \\ 0 \\ -2 \end{pmatrix}$

When $\lambda_2 = 1$ then $\mathbf{A} - \mathbf{I} = \begin{pmatrix} 3 & -3 & 3 \\ 6 & -6 & 3 \\ 0 & 0 & -3 \end{pmatrix}$

$$\begin{pmatrix} 3 & -3 & 3 \\ 6 & -6 & 3 \\ 0 & 0 & -3 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0$$

$$\Rightarrow 3x - 3y + 3z = 0 \quad (1)$$
$$6x - 6y + 3z = 0 \quad (2)$$

(2) $-$(1):

$$3x - 3y = 0$$

$\therefore x = y$

From (1): $3z = 0$

$\therefore z = 0$

$\therefore$ when $\lambda_2 = 1$, $\mathbf{v}_2 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$

$\therefore$ when $\lambda_1 = -2$, $\mathbf{v}_1 = \text{span}\left\{ \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} \right\}$, $\lambda_2 = 1$, $\mathbf{v}_2 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$

16 c)

$$\text{trace}(\mathbf{M}) = 4 - 5 - 2 = -3$$

17 a)

$$\begin{aligned}
\det \mathbf{M} &= \begin{vmatrix} 3a-b-c & -a+b & -a+c \\ 3a-2b-c & -a+2b & -a+c \\ 3a-b-2c & -a+b & -a+2c \end{vmatrix} \\
&= \begin{vmatrix} 3a-b-c & -a+b & -a+c \\ 3a-2b-c & -a+2b & -a+c \\ -c & 0 & c \end{vmatrix} \\
&= \begin{vmatrix} -c+2a & -a+b & -a+c \\ 0 & b & 0 \\ -c & 0 & c \end{vmatrix} \\
&= \begin{vmatrix} a & -a+b & -a+c \\ 0 & b & 0 \\ 0 & 0 & c \end{vmatrix} \\
&= a \begin{vmatrix} b & 0 \\ 0 & c \end{vmatrix} - 0 + 0 \\
&= abc
\end{aligned}$$

17 b)

$$\begin{pmatrix} 3a-b-c & -a+b & -a+c \\ 3a-2b-c & -a+2b & -a+c \\ 3a-b-2c & -a+b & -a+2c \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \lambda \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

$$\begin{aligned}
\mathbf{M} - \lambda \mathbf{I} &= \begin{pmatrix} 3a-b-c & -a+b & -a+c \\ 3a-2b-c & -a+2b & -a+c \\ 3a-b-2c & -a+b & -a+2c \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \\
&= \begin{pmatrix} 3a-b-c-\lambda & -a+b & -a+c \\ 3a-2b-c & -a+2b-\lambda & -a+c \\ 3a-b-2c & -a+b & -a+2c-\lambda \end{pmatrix}
\end{aligned}$$

$$\det(\mathbf{M} - \lambda \mathbf{I}) = \begin{vmatrix} 3a-b-c-\lambda & -a+b & -a+c \\ 3a-2b-c & -a+2b-\lambda & -a+c \\ 3a-b-2c & -a+b & -a+2c-\lambda \end{vmatrix} = 0$$

$$\begin{vmatrix} 3a-b-c-\lambda & -a+b & -a+c \\ 3a-2b-c & -a+2b-\lambda & -a+c \\ \lambda-c & 0 & c-\lambda \end{vmatrix} = 0$$

$$\begin{vmatrix} 3a-b-c-\lambda & -a+b & -a+c \\ \lambda-b & b-\lambda & 0 \\ \lambda-c & 0 & c-\lambda \end{vmatrix} = 0$$

$$\begin{vmatrix} 2a-b-\lambda & -a+b & -a+c \\ \lambda-b & b-\lambda & 0 \\ 0 & 0 & c-\lambda \end{vmatrix} = 0$$

$$\begin{vmatrix} a-\lambda & -a+b & -a+c \\ 0 & b-\lambda & 0 \\ 0 & 0 & c-\lambda \end{vmatrix} = 0$$

$$= (a-\lambda)\begin{vmatrix} b-\lambda & 0 \\ 0 & c-\lambda \end{vmatrix} - 0 + 0$$

$$= (a-\lambda)(b-\lambda)(c-\lambda)$$

So the eigenvalues are $\lambda = a$, $\lambda = b$ and $\lambda = c$

17 c)

$$3a-b-c-\lambda-a+2b-\lambda+c-\lambda = a+b+c$$

18 a)

$$\det\left(\frac{1}{2}\mathbf{M}\right) = \left(\frac{1}{2}\right)^3 \det(\mathbf{M})$$

$$\frac{1}{8}\begin{vmatrix} a+c & a-c & c-a \\ a-b & a+b & b-a \\ c-b & b-c & b+c \end{vmatrix}$$

$$= \frac{1}{8}\begin{vmatrix} a+c & a-c & 2c \\ a-b & a+b & 0 \\ c-b & b-c & 2c \end{vmatrix}$$

$$= \frac{1}{8}\begin{vmatrix} a+b & a-b & 0 \\ a-b & a+b & 0 \\ c-b & b-c & 2c \end{vmatrix}$$

$$= \frac{1}{8}\begin{vmatrix} 2b & 0 & 0 \\ a-b & 2a & 0 \\ c-b & 0 & 2c \end{vmatrix}$$

$$= \frac{1}{8}(2b)(2a)(2c) = abc$$

18 b)

$$\frac{1}{2}\begin{pmatrix} a+c & a-c & c-a \\ a-b & a+b & b-a \\ c-b & b-c & b+c \end{pmatrix}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \lambda\begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

$$\mathbf{M}-\lambda\mathbf{I} = \frac{1}{2}\begin{pmatrix} a+c & a-c & c-a \\ a-b & a+b & b-a \\ c-b & b-c & b+c \end{pmatrix} - \lambda\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$= \frac{1}{2}\begin{pmatrix} a+c-2\lambda & a-c & c-a \\ a-b & a+b-2\lambda & b-a \\ c-b & b-c & b+c-2\lambda \end{pmatrix}$$

$$\det(\mathbf{M}-\lambda\mathbf{I}) = \frac{1}{2}\begin{vmatrix} a+c-2\lambda & a-c & c-a \\ a-b & a+b-2\lambda & b-a \\ c-b & b-c & b+c-2\lambda \end{vmatrix} = 0$$

$$\begin{vmatrix} a+c-2\lambda & a-c & 0 \\ a-b & a+b-2\lambda & 2b-2\lambda \\ c-b & b-c & 2b-2\lambda \end{vmatrix} = 0$$

$$\begin{vmatrix} a+c-2\lambda & a-c & 0 \\ a-c & a+c-2\lambda & 0 \\ c-b & b-c & 2b-2\lambda \end{vmatrix} = 0$$

$$\begin{vmatrix} 2a-2\lambda & a-c & 0 \\ 2a-2\lambda & a+c-2\lambda & 0 \\ 0 & b-c & 2b-2\lambda \end{vmatrix} = 0$$

$$\begin{vmatrix} 2a-2\lambda & a-c & 0 \\ 0 & 2c-2\lambda & 0 \\ 0 & b-c & 2b-2\lambda \end{vmatrix} = 0$$

$$= (2a-2\lambda)(2c-2\lambda)(2b-2\lambda)$$

So the eigenvalues are $\lambda = a$, $\lambda = b$ and $\lambda = c$

18 c)

$$\text{trace}(\mathbf{M}) = \frac{1}{2}(a+c+a+b+b+c) = a+b+c$$